

CS215: Data Analysis and Interpretation

www.cse.iitb.ac.in/~sharat

(Many slides obtained from Prof. Ajit and Prof. Sunita and gratefully acknowledged. Fair use of internet resources as available to professors under copyright act)

And LLMs!!

1

Two questions from July 27

1. Could not understand what the course is about from yesterday's lecture.
2. Potential confusion on the example provided

2

First my checklist

- Announce piazza and Task00 due.
- Mentimeter results

3

What is the course about?

1. Could not understand what the course is about from yesterday's lecture.
2. I could give you the list of topics ... (next slide)

4

Topics

- Distributions and their properties
- Linear Regression
- Parameter Estimation
- Hypothesis Testing, Confidence Intervals
- Sampling, Multivariate Distributions
- Visualization?
- Others

5

What is the course about?

1. I could give you the list of topics ... (next slide)
2. But I decided to create a separate set of slides
 1. A sample!

6

Nature: The scientific publication

○ From [Wikipedia](#)

- **Nature** is a weekly international [scientific journal](#) publishing [peer-reviewed](#) research across the natural sciences, including biology, physics, chemistry, the earth sciences, and related interdisciplinary fields. *Nature* had one of the highest [impact factors](#) among multidisciplinary science journals (50.5), reflecting its strong citation influence within the scientific literature;^[1] some commentators also regard it as **among the most influential** scientific journals worldwide.^{[2][3][4]}
- In 2007, *Nature* (together with [Science](#)) received the [Prince of Asturias Award](#)
- As of 2012, it claimed an online readership of about three million unique readers per month.^[7]

7

What did Nature publish in 2003?

Adult persistence of head-turning asymmetry

A neonatal right-side preference makes a surprising romantic reappearance later in life.

A preference in humans for turning the head to the right, rather than to the left, during the final weeks of gestation and for the first six months after birth^{1,2} constitutes one of the earliest examples of behavioural asymmetry and is thought to influence the subsequent development of perceptual and motor preferences by increasing visual orientation to the right side^{3,4}. Here I show that twice as many adults turn their heads to the right as to the left when kissing, indicating that this head-motor bias persists into adulthood. My finding may be linked to other forms of sidedness (for example, favouring the right foot, ear or eye) that do not become established until long after the newborn head-turning preference has disappeared^{5,6}.

I observed kissing couples in public places (international airports, large railway stations, beaches and parks) in the United States, Germany and Turkey. The head-turning behaviour of each couple was recorded for a single kiss, with only the first kiss counted in instances of multiple kisses.

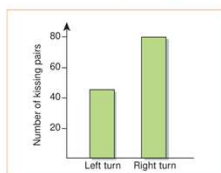


Figure 1 The number of couples who turn their heads to the right rather than to the left when kissing predominates by almost 2:1 (64.5%:35.5%; $n=124$ couples).

example, if the individual bias is also 2:1 to the right and if couplings are random, then four of nine pairs would be right kissers, one of nine would be left-kissing, and four of nine would be mixed; if choice is random in the last group (that is, two of these four pairs are right-turning and two are left-turning), the result for the nine couples would be a right-turn kissing bias in six of them. This neonatal 2:1 distribution for



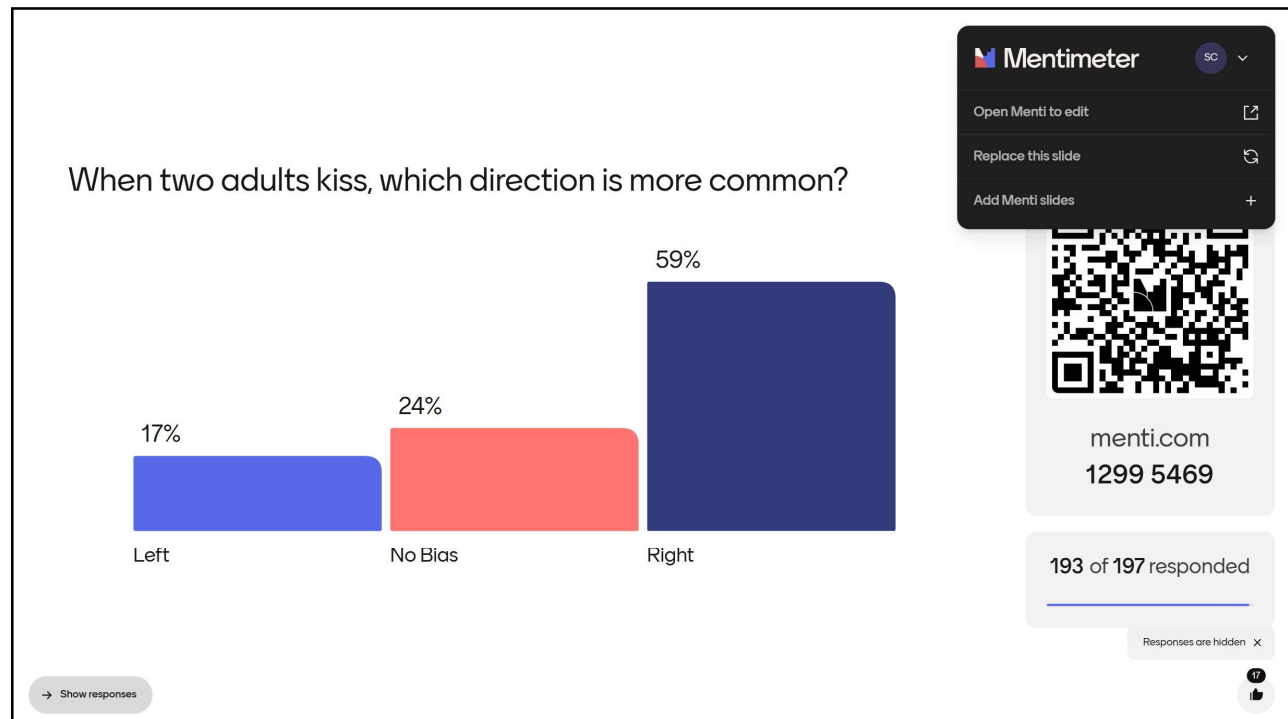
Figure 2 The Kiss, the couple in Antonio Canova's reproduction.

8

Audience poll

- You do not need experience to answer
- Just gut feeling is enough

9



10

Someone actually watched

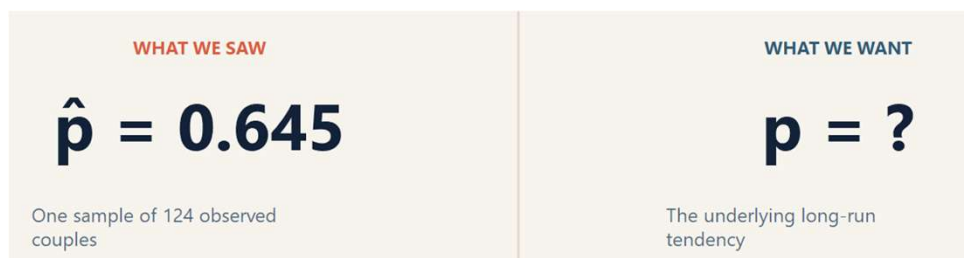
- Onur Gunturkun observed kissing couples in public places



11

Data != conclusion

- We expect 50%
 - Is that a signal?
 - Or chance?
- This course goes backward: data -> plausible truth



12

Statistics and Probability

- Same bridge. Opposite directions



13

Binomial distribution

- Right = 1
- Want $\hat{p}$ = average of the 0s and 1s = proportion turning right
- Proportion is a sample mean in disguise



14

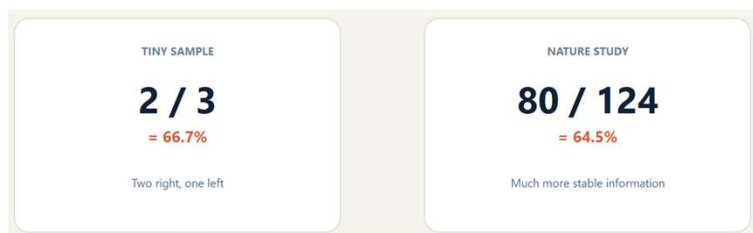
Inference has a price tag

- The model
 - Binary outcome
- Common tendency
 - Each observation shares on parameter p
- Independence
 - One observed couple does not determine the other

15

Point estimate

- A point estimate can be reliable
 - Or not
- Question for this course
 - How large is large enough?



16

Quantify uncertainty

- How surprising is 80 out of 124?
- Key starting point: No-bias benchmark of $p=0.5$

IF THERE WERE NO POPULATION-LEVEL BIAS

Expected right turns = 62

Observed = 80 • difference = 18

17

What is the course about?

Welcome to CS215!

18